Malachi Mabie

Software Engineer | Applied AI and Product Systems

Carpinteria, CA | Open to relocation to Los Angeles South Bay and Orange County

malachimabie.com | linkedin.com/in/mlmabie | github.com/mlmabie

Summary

Software engineer who owns customer systems from requirements through production operation. Current AMR work spans agents, robotics, and runtime integration. Shipped six systems for Drone Institute, led funded AI/ML research, and built software and tooling for semiconductor and medical manufacturing.

Experience

Advanced Metal Research / Gradient Industrial Robotics | July 2026 to Present

Robotics ML Research Engineer (contract)

- Build agent-assisted robotics software spanning motion-planning evaluation, typed task records, candidate trajectories, independent checks, and replayable experiment evidence.
- Develop Steam Deck operator controls and robot runtime integration. Resolved packet, identity, and configured-axis mismatches, contributing to operator-confirmed jogging on the assembled robot.

Mabie Industries, LLC | April 2025 to Present

AI Software Engineer / Independent Contractor

- Built and shipped six production systems for Drone Institute, including thermal imagery and GIS workflows, a marketplace, partner onboarding, operations, and order fulfillment. Owned architecture through deployment and production debugging.
- Built agent research infrastructure with typed tools, retrieval, memory, execution traces, replay, and independent verifiers. Delivered cognitive-security threat modeling and evaluation for Siethar Group.

RAM Laboratories, Inc. | April 2022 to March 2025

Senior Member of Technical Staff / Principal Investigator, AI/ML R&D

- Led T-UEBA, a $250,000 Army SBIR for tactical user and entity behavior analytics. Reported program evaluation results of 0.887 F1, under 3% false positives, and 15.2 ms per graph with a 15 MB temporal graph model.
- Directed VULCAN for the DARPA AI Cyber Challenge, combining code models with static analysis, fuzzing, and runtime verification to evaluate generated patches.

Endura Technologies | July 2019 to September 2020

Software Engineering and Deep Learning Internships

- Built Java chip-test instrumentation and Python automation; earlier work included audio classification with convolutional neural networks for TinyML.

Applied Medical | June 2013 to September 2018

Polymer Design and Mechanical Engineer; previously Process Development Intern

- Designed and qualified robotic end-of-arm tooling, pneumatic mechanisms, and manufacturing fixtures. A production insert-molding process fix reduced operating variance by approximately $100,000 per month.

Education

University of California San Diego | B.S. Cognitive Science, 2021

Specialization in Machine Learning and Neural Computation.

Technical skills

Python, TypeScript, JavaScript, SQL, FastAPI, PyTorch; APIs, agent tools, retrieval, memory, workflow state, evaluation, cloud operations; AWS, Docker, Linux.